

GAUTHAM BALASUBRAMANIAN

Senior Lead AI Engineer

9+ years | AI/ML | NLP | Computer Vision | LLMs | Agentic AI
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Canadian PR | Ontario, Canada

PROFILE SUMMARY

Results-driven Senior Lead AI Engineer with a Master's in Artificial Intelligence and 9+ years of experience across AI/ML, NLP, Computer Vision, LLMs, and Agentic AI. Proven track record of delivering end-to-end AI products at scale across startups and MNCs in Singapore and India. Specializes in building production-grade ML platforms, leading high-impact teams, and translating complex models into measurable business value.

EXPERIENCE

Staples India, Chennai *Apr 2025 – Present*

Senior Lead AI Engineer | Leading a team of 8

Unified Product Matching Platform

- Led the design and development of a Unified Matching Platform on Azure Databricks, enabling product matching across multiple business verticals through hybrid text search and an ensemble of image and text models (CLIP, MPNET, MARQO-B)
- Built an in-house re-ranker by fine-tuning QWEN-3, integrated into the platform's LLM-based reranking pipeline
- Replaced a legacy cross-referencing system, improving contract bid SLA by reducing turnaround time by **up to 0.6 days**, projected to generate **\$3M in EBITDA**
- Enhanced match quality and increased match coverage across verticals, projected to generate an **additional \$10M in revenue**
- Built an alternative product recommendation engine for the merchandising team leveraging internal matching infrastructure, projected to **increase RPV by 10%**
- Matched marketplace listings with internal catalog to power assortment intelligence, identifying gaps through comparisons with top-selling marketplace items to **drive Own Brand listing growth** and improve profit margins

Catalog Intelligence Platform

- Leading the build of a **Catalog Intelligence platform** that creates gold-standard product data, serving as a shared foundation for multiple use cases including digital experiences, search, agentic commerce, faster product onboarding to third-party marketplaces, and a hosted marketplace environment
- Building a proof-of-concept for **agentic catalog data enrichment** using Google's Agent Development Kit (ADK) and Gemini models
- Drove automated web-based item classification for accurate categorization and taxonomy alignment, accelerating product onboarding speed by **up to 40%**

U3 Infotech, Singapore *Jan 2024 – Jun 2024*

AI Engineer | AI Consultant for GIC Singapore | 6-Month Contract

- Developed LLM-powered analytics solutions to extract key metrics from financial documents
- Improved efficiency of financial report analysis **by 50%**, resulting in expedited decision making

UCARE.AI, Singapore *May 2021 – Dec 2023*

AI Scientist

Key Projects

- Developed a computer vision model for Orthokeratology treatment planning for Myopia using eye scan images; built data pipeline for custom ophthalmology equipment formats
- Developed models for insurance claim processing - an **overconsumption risk detection** model enabling Straight Through Processing (STP) with **80% efficiency improvement**, and a transformer-based pre-approval model delivering **70% faster payouts**

- Developed a LangChain + OpenAI LLM-based e-concierge for a major SEA insurer to handle queries and coverage requests
- Built end-to-end hospital bill estimation system deployed across leading Singapore hospital groups, reducing **error from 20% to 13%** and **improving range accuracy from 70% to 90%**
- Led POCs and pilots spanning medical image anonymization, hybrid RL-based health article recommendations, and diverse AI initiatives including chatbots, GANs, explainable AI, and synthetic data
- Drove team effectiveness through mentoring junior data scientists, sprint planning, code reviews, and developing reusable packages for automated reporting and dataset collection

NTT Data, Singapore *Sep 2020 – Apr 2021*

Senior Consultant | Data Science Consultant for DBS Singapore

- Built log analysis model for infrastructure maintenance, automated report generation, and developed REST APIs for internal operations

Tata Consultancy Services Ltd., Chennai *Nov 2015 – Jun 2019*

Systems Engineer

- Developed aircraft delay prediction model surpassing industry accuracy, published in IEEE, and contributed to multiple R&I proof-of-concepts across business verticals
- Built workforce management web application for a telecom client enabling demand forecasting and resource allocation

EDUCATION

National University of Singapore *Class of 2020*

Master of Technology in Intelligent Systems

Notable Projects: Conversational chatbot (Dialogflow + CLIPS + OptaPlanner) | Autonomous vehicles via RL (Markov Decision Process) + SLAM | UAV-based ground object detection using sensor fusion

AWARDS & PUBLICATIONS

- Published: "A Novel Approach: Airline Delay Prediction using Machine Learning" — IEEE Conference
- Awarded Best Team in Chennai — Tata Consultancy Services Ltd.
- Awarded Best Performer of the Year — twice during Bachelor's degree

SKILLS & TECHNOLOGIES

AI / ML: Machine Learning, Deep Learning, NLP, Computer Vision, Reinforcement Learning, Time Series Forecasting, Recommendation Systems, Embeddings, Reranking, Hybrid Search, Semantic Search

Agentic AI & LLMs: Agentic AI / Multi-Agent Systems, Google ADK, Gemini, MCP, LLMs, RAG, LLM Fine-tuning (LoRA/PEFT), LLM Evaluation & LLMOps

Frameworks: PyTorch, TensorFlow, Scikit-Learn, XGBoost, LightGBM, LangChain, Hugging Face, OpenCV, Spacy, YOLO

Data & Cloud: Azure Databricks, PySpark, Unity Catalog, Delta Lake, Databricks Workflows, Databricks AI Search (formerly Vector Search), Snowflake, Google BigQuery, Docker, Airflow, MLflow, ML-OPS, ClickHouse, Vector Databases, Graph Databases

Languages: Python, SQL, NoSQL, Shell Scripting, Git, Vue.js, Django, Flask

Visualization: Tableau, Power BI, Matplotlib, Seaborn, Plotly, Google Analytics